Sports Vision Assessment & Training System

Compact Product Requirements and Development Report

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Purpose: define a practical, compact first product for sports-vision assessment and training, including hardware, software, sport-specific protocols, algorithms, dashboards and validation.

1. Executive summary

This report defines a compact, modular sports-vision assessment and training product combining a small reaction-light kit, a display-based training application, optional camera tracking, and athlete-performance reporting. The first version should prioritise repeatable measurements and safe, usable training over a large number of unvalidated features.

Core principle: assessment and training are related but different. Assessment uses controlled, repeatable protocols to measure performance; training uses progressive exercises to practise a skill. Improvement on a device does not automatically prove improved performance in a real match. Keep measurement claims and training claims appropriately limited until validated.

2. Product scope and compact MVP

Recommended MVP: 8–12 addressable LED targets; one central controller; touch/proximity response sensing; a laptop or tablet; one standard monitor/TV; optional 60–120 fps camera; and a local-first application. Add eye tracking, VR, specialised visual-field equipment and advanced motion capture only after the MVP is reliable.

Primary users: sports vision/optometry practitioners, sports academies, coaches, sports scientists and athletes. The product should support supervised use and configurable protocols rather than presenting itself as a diagnostic medical device.

Initial sports: cricket, football, basketball, tennis, badminton, hockey, volleyball and boxing. Build a shared task engine with sport-specific profiles, rather than separate applications for every sport.

3. Visual needs to assess and train

- Simple reaction time: latency from a known visual stimulus to a defined response. Use a response button or sensor with measured timing behaviour.
- Choice reaction and response inhibition: select a response based on stimulus identity, or withhold a response for a no-go cue. Record speed and accuracy together.
- Peripheral detection and divided attention: detect a stimulus away from central fixation, with gaze/fixation requirements defined. A light-response task is not the same as a clinical visual-field examination.
- Visual tracking and pursuit: follow a moving target. Screen-based tracking can assess task performance, but accurate eye-movement metrics require an appropriate eye tracker.
- Saccadic orienting: shift gaze rapidly between targets. Do not claim saccadic latency from button-response time alone.
- Dynamic visual acuity and contrast sensitivity: assess ability to resolve moving or low-contrast targets using controlled optotypes/stimuli, viewing distance, luminance and calibrated procedures. These require more careful optical control than ordinary reaction lights.
- Depth perception and vergence/accommodation: assess with suitable validated tests and controlled viewing conditions; a flat display alone cannot fully assess real-world depth perception.
- Eye-hand and eye-foot coordination: relate visual stimulus to movement onset, movement time and target accuracy using a response surface, camera or instrumented target.
- Multiple-object tracking and visual attention: ask the athlete to track designated moving objects or identify relevant information among distractors. Record tracking accuracy, not only response speed.
- Anticipation and sport-specific decision-making: present representative video or simulation and ask the athlete to predict direction, trajectory or a suitable action. These tasks should be validated against the sport context.
- Baseline profile should document visual correction, relevant visual history, test conditions, handedness/footedness, sport, position, training experience, fatigue and any relevant environmental factors.

4. Compact hardware requirements

- LED targets: 8–12 units for MVP; individually addressable; at least 2–4 colours; diffuse, visible output; robust enclosure; stable mounting; optional touch/impact/proximity response sensing. Use a consistent response definition and verify sensor latency.
- Controller: ESP32-class MCU for a low-cost prototype, or a more deterministic controller if timing validation shows it is needed. Provide unique target IDs, firmware versioning, battery status and error reporting.
- Communication: BLE or Wi-Fi for commands/configuration; do not use the time a wireless message reaches the computer as the stimulus-onset timestamp. Timestamp stimulus onset as close to the hardware event as possible and synchronise clocks or calibrate offsets.
- Display: standard monitor/TV for MVP; record resolution, refresh rate, brightness and viewing distance. High-refresh displays are useful for motion tasks, but the true stimulus timing must be measured rather than assumed.
- Computer: modern laptop or mini PC, 8–16 GB RAM recommended, SSD, USB, Wi-Fi/Bluetooth. Tablet support may be added if the application and hardware SDK are stable.

- Camera: optional at MVP stage; 60 fps minimum for coarse movement timing, 120 fps or higher preferred for more detailed movement-onset analysis. A camera is not a substitute for eye tracking.
- Accessories: response button, tripod/stand, measuring tape, calibration card, charging equipment, carrying case and printed setup/safety guide.
- Safety and durability: rounded edges, secure mounting, strain relief, rechargeable protected batteries, temperature control, appropriate electrical/EMC testing, cleanable surfaces and a documented inspection procedure.

5. Compact software requirements

- Recommended architecture: local desktop/web application with a device service for hardware communication, task engine, data store and reporting module. A practical stack could be TypeScript/React or Angular for UI, Node.js/NestJS or Python FastAPI for service logic, SQLite for a single-station prototype, and PostgreSQL if multi-user/cloud deployment is required.
- Core modules: athlete and consent management; device setup/calibration; assessment protocols; training sessions; stimulus/response event logger; sport profiles; adaptive difficulty; results and trends; report export; role-based access; audit/error logs; data export and backup.
- Data model: Athlete, Session, ProtocolVersion, Trial, StimulusEvent, ResponseEvent, Device, CalibrationRecord, TrainingPlan and Report. Store raw trial-level data as well as derived summary metrics.
- Each trial should capture: athlete/session ID; protocol and version; stimulus type/position/colour; target eccentricity if applicable; scheduled and measured stimulus onset; response type and timestamp; correctness; missed/false response; movement onset if measured; device/firmware; display/camera settings; and environmental notes.
- Privacy: collect only necessary personal information, obtain appropriate consent, define retention and deletion procedures, encrypt data in transit and at rest where applicable, use role-based access, and avoid uploading identifiable video by default.

6. Core algorithms and metrics

- Reaction time (RT) = measured response timestamp – measured stimulus-onset timestamp. Report milliseconds and clearly define whether the endpoint is first sensor activation, button press, movement onset or movement completion.
- Simple reaction task: show a stimulus after a variable foreperiod; accept one defined response; reject anticipatory responses using a pre-set rule; record misses and false starts. Use several practice trials and enough scored trials to estimate a stable result.
- Choice reaction task: map stimulus categories to responses; calculate correct-trial median RT, accuracy, errors by category and missed trials. Do not hide errors by reporting speed alone.
- Go/no-go: respond to go cues and withhold for no-go cues; report hits, misses, false alarms, correct rejections and response latency.
- Peripheral task: present central fixation and peripheral cues at controlled positions; if fixation is essential, verify it with eye tracking or clearly label the task as peripheral-response training rather than validated fixation-controlled assessment.
- Tracking task: animate targets with known trajectories; ask the athlete to follow or identify target(s); score identification accuracy, target loss and response time. Use deterministic playback and consistent frame timing.
- Movement task: with synchronised camera/sensor timing, calculate movement-onset latency, movement duration, endpoint error and path efficiency. Define movement onset algorithmically and verify it against labelled video or a reference sensor.
- Summary statistics: median RT, mean RT where useful, standard deviation or robust variability, interquartile range, accuracy, omissions, false alarms and trial count. Predefine exclusion rules for anticipatory responses, hardware dropouts and invalid trials.
- Adaptive training: start at a baseline difficulty. Increase one dimension at a time (speed, number of targets, distractors, eccentricity or movement demand) when a pre-specified accuracy criterion is met; reduce difficulty if accuracy falls below a floor. Log all changes so sessions remain interpretable.
- Do not create a single 'vision score' until the component measures, weighting and validity are supported. Initially show separate skill metrics and a clearly labelled composite only if its calculation is transparent.

7. Dashboard requirements

- Coach/clinician overview: athlete, sport, position, latest session, adherence, key measures, safety/quality flags and progress trend.
- Athlete profile: age band, sport/position, dominant hand/foot, visual correction information if relevant, baseline results, goals and consent status.
- Session view: live task instructions, stimulus/response status, trial count, pause/stop controls, device connectivity, battery, calibration status and event-log warnings.
- Results view: median and mean RT; accuracy; false alarms; misses; variability; movement metrics; task settings; valid trial count; and comparison with the athlete's own baseline.
- Trend view: longitudinal charts by protocol version and conditions. Do not compare results from materially different devices, protocols or display settings without qualification.

- Sport profile view: training activities, skill focus, difficulty, session duration, progression rules and coach notes.
- Reporting: PDF summary plus CSV raw trial export. Include date, protocol/version, equipment configuration, test conditions, limitations and a statement that results are not a diagnosis.
- Usability: large high-contrast controls, simple setup wizard, clear athlete instructions, multilingual-ready text, keyboard/touch accessibility, and a visible emergency/stop control.

8. Sport-specific visual demands and training activities

Use the same core task engine, but select stimuli, response actions and progression according to sport and playing position. These are proposed training protocols, not validated normative tests.

8.1 Cricket

- Visual demands: ball tracking; pickup from bowler release; anticipation of line, length, swing and spin; hand-eye coordination; saccadic shifts; selective attention; field awareness.
- Assessment: direction/length prediction from standardised clips; moving-ball tracking; response choice; catch/target response latency; accuracy by ball type. Do not infer real batting skill from a screen task alone.
- Training: (1) track a moving ball and identify its final direction; (2) predict line/length from a release sequence; (3) choose leave/defend/attack in a scenario; (4) wicketkeeper/fielding response to random target direction; (5) add controlled distractors only after baseline accuracy is stable.
- Hardware: display or VR for trajectory tasks; LED targets and response button for choice reaction; camera only when movement measurement is needed.

8.2 Football / soccer

- Visual demands: peripheral awareness; scanning; tracking ball and players; spatial orientation; passing choice; anticipation; eye-foot coordination.
- Assessment: identify a teammate or open passing lane in a clip; peripheral target detection; left/right/front/back foot response; goalkeeper direction-choice task.
- Training: (1) respond with the correct foot to spatial cues; (2) scan a scene and identify the open player; (3) track the ball while detecting a peripheral cue; (4) predict pass/shot direction from a standardised video; (5) progress to dual-task movement drills under coach supervision.
- Hardware: LED targets around a safe training area, display/video module, optional floor markers or camera.

8.3 Basketball

- Visual demands: peripheral awareness; player and ball tracking; rapid passing decisions; near-far shifts; hand-eye coordination; selective attention.
- Assessment: identify a free teammate; track designated moving players; respond to colour/position cues; measure pass-target accuracy and latency.
- Training: (1) choose the correct passing target on screen; (2) track multiple players while identifying a designated player; (3) dribble or stationary ball-handling with a separate visual cue; (4) respond to a pass/shoot/hold decision; (5) progress to controlled movement and distractors.
- Hardware: display; LED targets; optional camera or instrumented ball/target. Screen-based tracking should not be claimed to measure actual game vision without validation.

8.4 Tennis

- Visual demands: ball tracking; serve-direction anticipation; rapid gaze shifts; depth and trajectory judgement; forehand/backhand response selection.
- Assessment: predict serve direction from standardised clips; track ball trajectory; respond to forehand/backhand cues; score direction accuracy and latency.
- Training: (1) forehand/backhand LED response; (2) predict serve direction before contact; (3) track a moving ball to its landing zone; (4) respond to variable ball speed and spin cues; (5) combine direction choice with a movement target.
- Hardware: LED targets; display/VR; optional camera for movement timing.

8.5 Badminton

- Visual demands: shuttle tracking; anticipation; rapid direction changes; eye-hand and eye-foot coordination; peripheral awareness.

- Assessment: identify shuttle trajectory from clips; directional footwork response; front/back/left/right choice reaction; landing-zone prediction.
- Training: (1) respond to four-corner targets; (2) predict smash/clear/drop direction; (3) track shuttle trajectory; (4) pair visual cues with controlled footwork; (5) add decision complexity gradually.
- Hardware: LED targets, display/VR, optional foot sensors.

8.6 Field hockey

- Visual demands: ball tracking at low height; peripheral scanning; teammate/opponent recognition; passing-lane selection; eye-hand/eye-body coordination.
- Assessment: find the ball or target player in a scene; choose a passing lane; respond to directional cues; measure decision accuracy.
- Training: (1) identify the ball among moving players; (2) select an open passing lane; (3) respond to left/right target cues while holding a stick safely; (4) track the ball while detecting peripheral movement; (5) progress to controlled stick-and-ball drills outdoors.
- Hardware: display/video; LED targets; optional camera.

8.7 Volleyball

- Visual demands: ball trajectory and landing prediction; reading setter/hitter cues; peripheral awareness; timing and hand-eye coordination.
- Assessment: predict ball landing zone from clips; identify attack direction; respond to block/defence cues; score accuracy and latency.
- Training: (1) choose a landing zone; (2) track a simulated serve/spike; (3) predict attack direction from body cues; (4) respond to front/back/left/right defensive positions; (5) add controlled distractors.
- Hardware: display/VR and LED targets.

8.8 Boxing

- Visual demands: detect opponent movement; recognise attack cues; response inhibition; peripheral awareness; rapid choice reaction.
- Assessment: identify a standardised attack direction from video; distinguish go/no-go cues; measure correct defensive-choice latency.
- Training: (1) respond to left/right/upper/lower target cues; (2) go/no-go response inhibition; (3) identify a defensive action from a clip; (4) progressively add combination recognition. Avoid unsafe, unexpected contact and use coach-supervised, non-contact protocols.
- Hardware: display, LED targets, response buttons; optional camera.

8.9 Other sports

- Table tennis: rapid ball-direction prediction and forehand/backhand choice tasks.
- Baseball: pitch tracking and swing/take decision tasks.
- Rugby: peripheral scanning, teammate recognition and passing choice.
- Archery/shooting: fixation stability and visual alignment only with suitable specialised equipment and qualified supervision.
- Swimming: start-signal response and visual orientation tasks; visual reaction scores should not be equated with total start performance.

9. Training session structure

Example 15–20 minute session: (1) equipment check and practice, 2 minutes; (2) baseline warm-up task, 2 minutes; (3) primary skill block, 5 minutes; (4) sport-specific decision block, 5 minutes; (5) cool-down/retest block, 2–4 minutes. Adjust to athlete needs and practitioner guidance.

Progress one variable at a time where possible. Use short blocks, breaks and a stop option if the athlete experiences discomfort, dizziness, headache or visual strain. Do not use strobe or VR modules without suitable safety screening and supervision.

10. Validation and quality assurance process

- Stage 1 — bench verification: measure LED/display onset and sensor response timing with a photodiode, oscilloscope or suitable reference device. Repeat across targets, colours, battery levels and communication conditions. Document timing error and variability.

- Stage 2 — software verification: test trial sequencing, randomisation, missed responses, false starts, duplicate events, dropped messages, clock offsets, pause/resume, device disconnection and data export. Preserve raw logs.
- Stage 3 — usability pilot: test setup, instructions, comfort, session duration and error recovery with representative coaches and athletes. Revise confusing tasks before efficacy testing.
- Stage 4 — test–retest reliability: repeat the same protocol under comparable conditions, ideally with an appropriate interval and practice trials. Estimate ICC or other suitable reliability statistics, standard error of measurement and minimal detectable change where appropriate.
- Stage 5 — validity: compare each intended measure with an appropriate reference method. For reaction time, use a validated timing reference; for eye movements, use an eye tracker; for visual acuity/contrast, use calibrated validated procedures. A simple light board cannot validate every visual function.
- Stage 6 — learning-effect analysis: include familiarisation, standardised instructions and enough trials; examine whether score changes reflect task learning rather than underlying ability.
- Stage 7 — training transfer study: where feasible, compare a training group with a control or comparison group, pre-register outcomes, use sport-relevant performance measures and blinded scoring where possible. Report effect sizes, uncertainty and limitations.
- Stage 8 — release control: version protocols and firmware, maintain calibration records, establish acceptance thresholds before data collection, document bugs and corrective actions, and revalidate material hardware/software changes.

Report claims carefully: separate device reliability, improvement on the trained task, and transfer to competitive performance. Do not claim diagnosis, injury prevention or improved match performance without appropriate evidence and regulatory review.

11. Safety, privacy and regulatory considerations

- Obtain appropriate consent and parental/guardian consent where required for minors. Define who can access athlete records and video, and how data are stored, exported and deleted.
- Use secure mounting and protected batteries; inspect devices before each session. Maintain a clear stop/pause control. Provide rest breaks and stop if the athlete experiences discomfort.
- Screen and supervise higher-risk activities such as stroboscopic eyewear and VR; they may be unsuitable for some users. Do not use them while moving near obstacles or performing contact drills.
- If marketed for clinical diagnosis or treatment, seek regulatory advice on applicable Indian medical-device requirements and relevant electrical, EMC, software and data-protection obligations. Avoid diagnostic claims until the appropriate pathway is completed.

12. Suggested implementation roadmap

- Weeks 1–2: confirm intended use, sport priorities, protocols and acceptance criteria; create a requirements traceability matrix.
- Weeks 3–6: build 8–12 LED prototype, controller firmware, response sensing and bench timing test.
- Weeks 5–9: build athlete/session workflow, trial engine, event logging, results dashboard and CSV/PDF export.
- Weeks 8–12: implement simple reaction, choice reaction, go/no-go, peripheral-response training and visual-search tasks.
- Weeks 10–14: add two initial sport profiles (recommended cricket and football, or the sports your pilot partner prioritises).
- Weeks 14–18: conduct bench verification and a small usability/reliability pilot; fix timing and workflow problems before adding advanced modules.

These are indicative planning ranges for a small team, not guaranteed delivery dates. Add eye tracking, VR and clinical visual-function testing only after the core system meets its acceptance criteria.

13. MVP acceptance checklist

- All trials record protocol version, stimulus onset, response timestamp, response type and correctness.
- Stimulus timing and response-sensor latency are independently measured and documented.
- Invalid trials and hardware failures are flagged, not silently discarded.
- Median reaction time, variability, accuracy, omissions and false alarms are calculated correctly.
- Same protocol can be repeated with matching settings and exported raw data.
- Dashboard shows baseline and longitudinal results without unsupported normative rankings.
- Sport profiles clearly identify visual demand, task, response, scoring rule and progression rule.
- User consent, role access, backup/export and data-deletion processes are defined.
- Safety instructions, calibration procedure, troubleshooting guide and version history are available.

14. Key recommendation

Build a compact, reliable platform first: 8–12 LED targets + response sensing + display-based tasks + athlete dashboard + raw data export. Focus on timing validity, repeatability, ease of use and two well-designed sports profiles before expanding to all sports.

Treat eye tracking, dynamic visual acuity, clinical visual-field testing, VR and AI-based movement analysis as separate advanced modules, each requiring its own hardware and validation. This staged approach reduces cost and avoids overstating what the first prototype can measure.